Part 16 — Nine more problems: separations, quantitative omics, consortia

The three categories that part 13 gave theory to and no problems. Same format as X1–X24: statement, stated assumptions, numbered questions, worked solution with every figure computed in this session. Together with the 56 in your appendix and the 30 in part 15, this takes the set to **95**.

Group	Problems	Extends
Downstream separations	Z1–Z4	Chapter S1, weeks 44 and 49
Quantitative proteomics and metabolomics	Z5–Z6, Z9	Chapter S5, weeks 43 and 51
Consortia dynamics	Z7–Z8	Chapter S4, weeks 9 and 26–27

Group A — Downstream separations

Z1 · extends chapter S1 · 2 h

Breakthrough, dynamic capacity, and the bed you are not using

A capture column holds 100 L of resin in a 200 mm bed. The resin's static binding capacity is 65 g/L; at 10% breakthrough and a 2-minute residence time its dynamic capacity is 45 g/L. Feed titer is 10 g/L and the campaign must process 500 kg of product.

1. Compute the fraction of the bed that is doing no work, and express it as a length of unused bed.
2. Dynamic capacity rises toward the static value as residence time increases, roughly as SBC$(1 - e^{-t/2.5})$ with t in minutes. Tabulate capacity and the number of cycles at residence times of 2, 4 and 6 minutes.
3. Doubling the residence time doubles the loading time per cycle. Does it reduce total campaign time?
4. What would you measure to get the real curve, and what does 10% breakthrough mean for product loss?

Worked solution. DBC/SBC = 45/65 = 0.69, so **31% of the bed is unused** — the mass-transfer zone that has not yet saturated when breakthrough begins — which is **62 mm of the 200 mm bed**. That length is the quantity a column designer actually works with: it is roughly independent of bed height, so a taller bed wastes a smaller *fraction*, which is why capture columns are tall and narrow rather than short and wide.

Residence time	Dynamic capacity	Load per cycle	Cycles for 500 kg
2 min	35.8 g/L	3.6 kg	140
4 min	51.9 g/L	5.2 kg	96
6 min	59.1 g/L	5.9 kg	85

Doubling residence from 2 to 4 minutes raises capacity by **45%** and cuts cycles from 140 to 96 — a 31% reduction — while doubling the load time per cycle. Since loading is only part of a cycle (equilibration, wash, elution, regeneration are the rest), the net effect is usually favourable, and the answer is obtained by writing the full cycle time rather than by intuition. The measurement to make is a **breakthrough curve at each candidate flow rate**: load at constant concentration and plot outlet against loaded mass; DBC is read at your chosen breakthrough, and the curve's steepness gives the mass-transfer zone directly. And 10% breakthrough means exactly that — **at the end of loading you are losing 10% of the feed to the flow-through**, so the capacity figure and the yield figure are two readings of the same curve.

What it teaches: dynamic capacity is a kinetic quantity, not a property of the resin, and the breakthrough criterion you choose sets both the capacity and the product you throw away.

Z2 · extends chapter S1 · 1.5 h

A three-step train: where purity comes from and what it costs

A secreted protein at 35% purity in clarified broth must reach 98%. The available steps are centrifugation plus depth filtration (93% yield, purifies 2-fold), ion-exchange capture (90% yield, 8-fold) and hydrophobic-interaction polish (88% yield, 22-fold), where "n-fold purification" means the impurity fraction falls n-fold.

1. Track purity and cumulative recovery through the train.
2. Is the 98% specification met, and with what margin?
3. What do one and two further polish steps buy, and cost?
4. How does recovery enter the cost per kilogram delivered?

Step	Yield	Purity after	Cumulative recovery
Clarification	0.93	67.5%	93.0%
Ion-exchange capture	0.90	95.9%	83.7%
HIC polish	0.88	**99.82%**	**73.7%**

Worked solution. The specification is met with room to spare — 99.82% against 98% — at an overall recovery of **73.7%**. Adding a fourth step takes purity to 99.97% and drops recovery to **66.3%**; a fifth gives 99.99% at **59.7%**. So each additional polish buys a decimal place of purity and costs roughly seven points of recovery, which is the quantitative form of chapter S1's argument that the train should be as short as the specification allows. Because the

product delivered is what you sell, the cost per kilogram scales as 1/recovery: at 73.7% it is **1.36 times** the single-pass cost of the materials and operations, and at 59.7% it is 1.68 times.

The design judgement is to *stop*: a group that habitually adds a final polish "for safety" is paying 7% of its product for purity nobody specified. The time to decide is when the specification is written, and the number to write beside it is the recovery the train can deliver.

What it teaches: purity and recovery trade against each other step by step, and over-purifying is a silent, recurring cost.

Z3 · extends chapter S1 · 1.5 h

Crystallisation: read the yield off the solubility curve

A product's solubility follows $c^* = 3.0 \cdot e^{0.035T}$ g/L with T in °C. The crystalliser is charged at 60 °C with a saturated solution.

1. Tabulate solubility at 60, 40, 25, 10 and 4 °C.
2. Compute the crystallisation yield for cooling from 60 °C to 25, 10 and 4 °C.
3. What does a 1.5-fold concentration step before cooling add?
4. With a metastable zone width of 6 °C, what constrains the cooling profile, and why is a linear ramp a poor choice?

Temperature	60 °C	40 °C	25 °C	10 °C	4 °C
Solubility	24.5 g/L	12.2	7.2	4.3	3.5

Worked solution. Yield is $(C_1 - C_2)/C_1$, so cooling to 25 °C gives **70.6%**, to 10 °C **82.6%** and to 4 °C **85.9%** — the last 6 °C of cooling buys only 3.3 points because the solubility curve is flattening, and it costs refrigeration. Concentrating 1.5-fold first raises the 10 °C yield to **88.4%**, which is the better lever: evaporation is cheaper than refrigeration per unit of yield here, and it also shrinks the vessel.

The metastable zone is what constrains the profile. Supersaturation must stay inside it — beyond the width, primary nucleation produces a shower of fines that will not filter. Since the yield is driven by the solubility *difference*, and that difference grows as the solution cools, a **linear temperature ramp generates supersaturation fastest at the end**, exactly when the crystal surface area is largest but the driving force is highest; it spends most of the batch near the metastable limit. The standard answer is to seed just below saturation and cool slowly at first, then faster — a cubic profile — so that supersaturation is held roughly constant as the growing crystal area consumes it.

What it teaches: crystallisation yield is pure thermodynamics off the solubility curve, while crystal quality is entirely a kinetic question about the cooling path.

Z4 · extends chapter S1 · 1.5 h

Diafiltration: diavolumes, sieving and what leaks through

An ultrafiltration step washes an impurity out of a 20 m³ retentate. The impurity passes the membrane with sieving coefficient S_i and the product with S_p; in N diavolumes the impurity falls as $e^{-S_i N}$ and the product is retained as $e^{-S_p N}$.

1. How many diavolumes for 10-, 100- and 1,000-fold impurity clearance with perfect sieving?
2. For four membrane cases, compute the diavolumes needed for 100-fold clearance and the product retained.
3. Compute the buffer volume and the time for 7 diavolumes at 2 m³/h of permeate.
4. A two-stage cascade recycles permeate. What does recovering 50% or 80% of the lost product do?

Worked solution. With perfect sieving the requirement is ln F: **2.3, 4.6 and 6.9 diavolumes**. The membrane cases show where the design actually lives:

S_i (impurity)	S_p (product)	Diavolumes for 100×	Product retained
1.0	0.00	4.6	100%
1.0	0.02	4.6	91.2%
0.8	0.05	5.8	75.0%
0.6	0.10	7.7	**46.4%**

The two coefficients compound viciously: a membrane that partly retains the impurity needs more diavolumes, and every extra diavolume loses more product, so the worst case keeps **under half** of what it started with. Buffer for 7

diavolumes on 20 m³ is **140 m³**, which at 2 m³/h of permeate takes **70 hours** — so the step is also the schedule bottleneck, and in a sterile process that is 70 hours of hold time to validate.

Recycling the permeate recovers part of the loss: at S_p = 0.10 and 7.7 diavolumes, retention rises from 46.4% to **73.2%** with 50% permeate recovery and **89.3%** with 80%. That is the argument for a cascade, and the cost is a second membrane stage and more complexity in a step that is already the longest in the train.

What it teaches: a membrane is specified by two sieving coefficients, not one, and the product you lose is set by the diavolumes the impurity's coefficient forces you to run.

Group B — Quantitative proteomics and metabolomics

Z5 · extends chapter S5 · 2 h

Absolute protein quantification, and the flux bound it feeds

You spike known amounts of isotope-labelled AQUA peptides and quantify three peptides per protein from their light-to-heavy ratios, with 5% or 15% CV on each ratio. The protein is a 50 kDa enzyme and the cell carries 0.5 g of protein per gram dry weight.

1. Recover the five spiked amounts at both CVs and report the mean absolute error.
2. Compute the protein-level CV from three peptides, and say why the median of three is used rather than the mean.
3. Convert abundance to a flux bound $v \le k_{cat}[E]$ for proteome fractions of 0.1, 1 and 5% and k_{cat} of 1, 10 and 100 s^{-1}.
4. State the uncertainty on the bound, and what the bound is still good for.

Worked solution. Recovery across 0.5 to 120 fmol has a mean absolute error of **7.3% at 5% ratio CV and 10.0% at 15%** — the method is accurate because the labelled standard is chemically identical, so the error is close to the ratio precision itself. Averaging three peptides gives a protein-level CV of $0.15/\sqrt{3}$ = **8.7%**, but the reason for taking the **median** rather than the mean is bias, not variance: a peptide with a missed cleavage or a partial modification reads systematically low — by 5%, 15% or 30% in the cases tabulated — and one bad peptide of three shifts a mean while leaving a median untouched.

Proteome fraction	k_{cat} = 1 s^{-1}	10 s^{-1}	100 s^{-1}
0.1%	0.04	0.4	3.6
1%	0.4	3.6	36.0
5%	1.8	18.0	180.0

Bounds are in mmol·gDW^{-1}·h^{-1}. The uncertainty is the product of the two errors and is dominated by k_{cat}: a literature value wrong by tenfold makes the bound wrong by **tenfold**, while a 25% abundance CV contributes 25%. So the bound is good to **about a factor of ten**. It is still worth computing, because the useful statement is an exclusion: a design needing 36 mmol·gDW^{-1}·h^{-1} from an enzyme with k_{cat} = 1 s^{-1} would need 100% of the proteome, and no plausible k_{cat} error rescues that.

What it teaches: absolute quantification is accurate and the flux bound built on it is not, because the k_{cat} dominates — so report bounds as exclusions with their assumed k_{cat} named.

Z6 · extends chapter S5 · 1.5 h

Quenching and extraction: choose your artefact deliberately

Four sampling methods are compared for intracellular metabolites, with their reported recoveries and whether they separate intracellular from extracellular pools.

1. Tabulate the four methods against recovery and separability, and say what each artefact does to a measurement.
2. Your product accumulates to twenty times its intracellular concentration in the broth. What fraction of an unwashed "intracellular" measurement is artefact?
3. Compute pool turnover times for four pool-and-flux combinations and the fraction lost in a 2-second quench.
4. Which method would you choose for a thermodynamic analysis, and what would you report?

Method	ATP recovery	Glucose-6-P	Separates pools?
Cold methanol, washed	0.45	0.60	yes
Cold methanol, unwashed	0.95	0.97	**no**
Fast filtration under nitrogen	0.80	0.85	yes
Whole broth, boiling ethanol	0.98	0.99	**no**

Worked solution. The trade is exact and unavoidable: washing separates the pools and leaks the metabolites, not washing retains them and measures both pools together. With the product twenty times more concentrated outside than inside, an unwashed measurement is **95% extracellular** — it is not a measurement of the cell at all. Turnover is the

other constraint: a 0.05 mM pool at 2 mM/s turns over in **0.03 s**, and a 2-second quench loses essentially all of it; only the 5 mM pool at 0.3 mM/s, turning over in 17 s, survives with 12% loss.

For a thermodynamic analysis the right choice is **fast filtration under nitrogen** — it separates the pools and recovers 80–85% — with two things reported alongside: the recovery measured per compound by a second spike, and the extracellular concentration measured separately so the intracellular figure can be corrected rather than assumed. And the honest statement in the methods is that no method is artefact-free; what makes the number usable is that the artefact was quantified.

What it teaches: every sampling method trades leakage against separability, and the pool you most want to measure is the one that turns over fastest.

Z9 · extends chapter S5 · 1.5 h

Instrument drift, and the pooled QC that catches it

A metabolomics batch of 120 injections loses 30% of its signal across the run as the source fouls. A pooled quality-control sample is injected every tenth position.

1. Compute the coefficient of variation of the QC injections before correction.
2. Correct the run by interpolating the QC signal and recompute the QC CV and the residual bias between the first and last ten samples.
3. What happens to a comparison between two conditions if the samples were injected in condition order?
4. State the three rules this implies for a batch design.

Worked solution. The uncorrected QC injections have a CV of **12.4%** — entirely drift, since the QC sample is the same material every time. Interpolating the QC trace and dividing it out brings the QC CV to **essentially zero** by construction and leaves a residual bias of **1.1%** between the first and last ten samples, which is the part the interpolation could not capture.

If the samples were injected in condition order — all controls then all treatments, as is natural when they come off the bench that way — the 30% drift maps exactly onto the comparison and appears as a **30% treatment effect**. Nothing in the data distinguishes it from biology, and no statistical test helps, because the confounding is in the design rather than the noise. Hence the three rules: **randomise the injection order** across conditions, **inject a pooled QC every ten samples** and correct against it, and **report the QC CV with the data** so a reader can judge the correction. A feature whose QC CV exceeds about 20–30% should not be interpreted at all.

What it teaches: instrument drift becomes a treatment effect whenever injection order follows condition, and the pooled QC is the only thing that separates them.

Group C — Consortia dynamics

Z7 · extends chapter S4 · 2 h

Designing the population ratio you actually want

Strain A consumes glucose and excretes α moles of an intermediate per mole consumed; strain B grows only on the intermediate. In a chemostat at D = 0.2 h^{-1} with S_0 = 20 g/L, both strains have μ_{max} = 0.5 h^{-1} and Y = 0.4 g/g, with K_S = 1.0 and K_I = 0.5 g/L.

1. Compute the steady state and the A:B ratio for the base case.
2. Vary α to 1.0 and 4.0, Y_B to 0.2, $\mu_{max,B}$ to 0.3 and D to 0.35. Tabulate the ratio.
3. Derive the steady-state intermediate concentration analytically and check it against the simulations.
4. You need a 1:1 ratio. Which parameter do you engineer, and how would you do it?

Case	A (g/L)	B (g/L)	A:B	Intermediate
Base	7.73	6.05	1.28	0.333 g/L
α = 1.0	7.73	2.96	2.61	0.333
α = 4.0	7.73	12.24	0.63	0.333
Y_B = 0.2	7.73	3.03	2.56	0.333
$\mu_{max,B}$ = 0.3	7.73	5.79	1.34	1.000
D = 0.35	7.07	5.19	1.36	1.167

Worked solution. The ratio responds to **α and Y_B and almost nothing else**: halving α doubles the ratio, and halving Y_B does the same, while changing B's growth rate barely moves it. The reason is in the third question's algebra. At steady state B grows at D, so $\mu_{max,B} I/(K_I + I) = D$, giving

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I* = K_I·D/(μ_max,B − D)
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which is the chemostat result applied to the intermediate — **0.5 × 0.2/0.3 = 0.333 g/L** for the base case, **0.5 × 0.2/0.1 = 1.000** when $\mu_{max,B}$ falls to 0.3, and **0.5 × 0.35/0.15 = 1.167** at D = 0.35, matching all three simulations exactly. B's kinetics set the intermediate *concentration*; the stoichiometry and yields set the *biomass ratio*. That separation is the design insight: the two are controlled by different parameters, so you can specify both.

To force 1:1 from the base case's 1.28 you raise B's biomass per unit of intermediate or lower A's excretion — practically, you tune α by changing the export or the branch flux in A (part 3 chapter 45), or you accept the ratio and tune the *inoculum of the subsequent batch*, which does nothing at steady state but matters in a batch that never reaches one.

What it teaches: in an obligate cross-feeding consortium the intermediate concentration and the population ratio are set by different parameters, and only the ratio is a design variable you can hit.

Z8 · extends chapter S4 · 1.5 h

How long before the cheater wins?

A defector that stops paying its share grows a fraction s faster. Starting from one part in a million in a chemostat at D = 0.2 h^{-1}, the defector's frequency follows the logistic replacement $p(t)/(1 - p(t)) = p_0/(1 - p_0)\cdot e^{sDt}$.

1. Compute the time to reach 50% and 99% of the population for advantages of 2, 5, 8 and 15%.
2. Express those as generations.
3. Compare with a 7-day batch and a 60-day continuous campaign.
4. Under reciprocal auxotrophy the defector cannot grow at all. How fast does it wash out at D = 0.05 and 0.2 h^{-1}?

Advantage s	To 50%	To 99%	Generations to 99%
2%	3,454 h	4,603 h	1,328
5%	1,382 h	1,841 h	531
8%	863 h	1,151 h	332

15%	461 h	614 h	177

Worked solution. Against a 7-day batch (168 h) every case is safe — a cheater starting at 10^{-6} cannot sweep in a week at any plausible advantage. Against a 60-day continuous run (1,440 h) the picture reverses: at an 8% advantage the defector is past 99% before the campaign ends, and at 15% it takes over in the first month. **The decisive variable is campaign length, not the advantage**, which is why batch processes tolerate designs that continuous processes cannot.

Reciprocal auxotrophy changes the kind of problem rather than its size. A strain that cannot grow has $\mu = 0$ and is simply diluted out at the dilution rate: **tenfold in 46 h and a thousandfold in 138 h at $D = 0.05$**, and **12 h and 35 h at $D = 0.2$**. So the defence is not that cheating is slow but that cheating is lethal — the frequency moves in the opposite direction, and faster at higher dilution. The caveat worth stating is that this holds only while the medium lacks the complemented nutrients; a rich feed or a lysing population restores cross-feeding and the protection with it.

What it teaches: a cheater's threat is set by the run length against its advantage, and reciprocal auxotrophy inverts the dynamics rather than slowing them.

Closing notes to part 16

The nine in one page

Z1 31% of a capture bed does no work at breakthrough, which is 62 mm of a 200 mm bed; doubling residence time raises capacity 45% and cuts cycles from 140 to 96. **Z2** a three-step train reaches 99.8% purity at 73.7% recovery, and each further polish buys a decimal place for seven points of recovery. **Z3** cooling 60 → 10 °C crystallises 82.6%, and a 1.5-fold concentration step beats the last 6 °C of refrigeration. **Z4** a membrane with $S_i = 0.6$ and $S_p = 0.10$ needs 7.7 diavolumes and keeps under half the product; permeate recycle restores it to 73–89%.

Z5 AQUA recovers spiked amounts to 7–10%, but the flux bound built on it is good only to a factor of ten because k_{cat} dominates. **Z6** an unwashed intracellular sample is 95% artefact when the product is 20-fold concentrated outside, and a 0.05 mM pool turns over in 0.03 s. **Z9** a 30% drift across 120 injections becomes a 30% treatment effect if injection order follows condition.

Z7 the intermediate concentration is $K_I D/(\mu_{max,B} - D)$, set by the consumer's kinetics, while the population ratio is set by α and the yields — two parameters, two controls. **Z8** a cheater at 8% advantage sweeps a 60-day campaign and cannot touch a 7-day batch; under reciprocal auxotrophy it washes out a thousandfold in 35 h.

On the numbers

All computed in this session. The assumptions worth challenging: the empirical capacity-against-residence form in Z1, which should be replaced by your own breakthrough curve; the fold-purification figures in Z2, which are illustrative and step-specific; the solubility expression in Z3; the recovery figures in Z6's method table, which are representative of the literature rather than measured here and are the kind of thing sheet 10 of the lookup workbook exists for; and the logistic replacement in Z8, which assumes a constant advantage and well-mixed conditions.

Where the problem set now stands

Source	Problems	Data
Your curriculum appendix, A1–W3	56	Stated inline; 7 quote a measured parameter from a paper
Part 15, X1–X24	24	Computed this session; simulated data with known truth where needed
Part 15, Y1–Y6	6	Real public datasets, retrieval routes given, **not run**
Part 16, Z1–Z9	9	Computed this session
Total	**95**	

Coverage against the nine skill categories is now even, with the thinnest remaining areas being genome-scale model curation, which needs software I cannot run, and anything depending on a real instrument's output. Both are better done at a bench or a terminal than on paper.